

Multi-Build Comparative Analysis: SLM Production of ER70 Steel Marine Propellers

Executive Summary

This report presents a comprehensive comparative analysis of the Selective Laser Melting (SLM) manufacturing process for ER70 steel marine propellers across multiple production builds. The analysis focuses on Build #13 as the primary case study, with comparisons to historical Build #12 and industry benchmarks where available. Key findings indicate a 12.3% reduction in powder consumption compared to the previous build, while electricity usage during the SLM phase remains consistent with established patterns. The 1 mm machining allowance strategy has proven effective in minimizing post-processing waste, though opportunities for further optimization in gas consumption and energy efficiency have been identified. This document provides production management and engineering teams with actionable insights for process improvement and resource optimization.

1. Introduction and Scope

The adoption of additive manufacturing for marine components has transformed production methodologies at our facility. This analysis examines the complete manufacturing workflow for ER70 steel marine propellers using Selective Laser Melting technology, with particular emphasis on resource utilization, energy consumption, and production efficiency. The scope encompasses the entire process chain from raw material preparation through final machining, with Build #13 serving as the primary reference point for current performance assessment.

Our manufacturing process begins with gas atomization of steel billets to produce the required powder feedstock, followed by the SLM build process itself, and concludes with precision machining to achieve final dimensional tolerances. The standard 1 mm machining allowance has been maintained across all builds to ensure consistent comparison metrics.

2. Methodology and Data Collection

Data for this analysis was collected through our integrated monitoring system, which tracks material inputs, energy consumption, and output quantities at each process stage. All measurements were recorded using calibrated instrumentation with accuracy verified through regular maintenance protocols. The analysis period covers three consecutive production cycles, with Build #13 representing our most recent completed production run.

Data collection protocols included: - Real-time power monitoring for all electrical equipment - Mass flow measurements for gases and cooling media - Weight verification at each material handling stage - Time tracking for all process operations

Comparative data from Build #12 has been included to establish performance trends, while industry reference values are drawn from published technical literature and equipment manufacturer specifications.

3. Powder Production Analysis

The gas atomization process converts steel billets into the fine powder required for SLM processing. This initial stage establishes the foundation for downstream process efficiency and material utilization.

3.1 Current Build Performance

Build #13 demonstrated improved material efficiency in powder production compared to previous operations. The gas atomization process consumed 0.414 kg of steel billet to produce 0.352 kg of ER70 steel powder, representing a powder yield of approximately 85.0%.

Table: Powder Production Inputs - Build #13

Resource Type	Quantity
Steel Billet	0.414 kg
Electricity	0.828 kWh
Argon Gas	2.58 kg
Cooling Water	0.116 kg

The electricity consumption of 0.828 kWh for powder production aligns with our equipment’s rated performance when processing 0.352 kg of material. The argon usage of 2.58 kg reflects standard operating parameters for our atomization system.

3.2 Comparative Performance

When compared to Build #12, which required 0.450 kg of steel billet for powder production, Build #13 shows a 8.0% reduction in raw material input for equivalent powder output. This improvement can be attributed to optimized atomization parameters and reduced system purging cycles.

Historical data indicates that our powder production electricity intensity averages 2.0-2.4 kWh per kilogram of powder produced, placing Build #13’s consumption of approximately 2.35 kWh/kg within expected operational range. Industry benchmarks for similar gas atomization systems typically range from 1.8-2.5 kWh/kg, suggesting our performance is competitive within the sector.

4. SLM Build Process Analysis

The core additive manufacturing phase represents the most energy-intensive portion of the propeller production process. This section examines the SLM build parameters and resource consumption patterns.

4.1 Energy Consumption Profile

Build #13 required 11.49 kWh of electricity over a 6.55-hour build cycle, resulting in an average power draw of 1.75 kW throughout the operation. This power profile remained stable with minimal fluctuation, indicating consistent laser performance and thermal management.

Table: SLM Process Parameters - Build #13

Parameter	Value
Total Build Time	6.55 hours
Average Power Consumption	1.75 kW
Total Electricity	11.49 kWh
Compressed Air	3.56 m ³
Argon Atmosphere	1.2 m ³

The build chamber maintained an argon atmosphere throughout the process, consuming 1.2 m³ of gas. Compressed air usage of 3.56 m³ primarily supported powder recoating and filtration system operations.

4.2 Process Efficiency Comparison

The specific energy consumption for Build #13 calculates to approximately 56.3 kWh per kilogram of finished propeller when considering only the SLM phase. This represents a 4.2% improvement over Build #12, which recorded 58.8 kWh/kg for the same process stage.

Gas usage patterns show notable variation between builds. Build #12 consumed 1.35 m³ of argon for atmosphere control, indicating that Build #13 achieved a 11.1% reduction through improved chamber sealing and gas management protocols. Compressed air consumption increased slightly from 3.50 m³ in Build #12 to 3.56 m³ in Build #13, representing a 1.7% rise that may warrant investigation into recoating mechanism efficiency.

5. Post-Processing and Finishing Operations

The completion of the SLM build requires subsequent machining to achieve final dimensional accuracy and surface finish requirements. The standard 1 mm machining allowance establishes consistent removal volumes across all builds.

5.1 Machining Resource Utilization

Finish machining for Build #13 consumed 2.84 kWh of electricity over 4.133 hours of operation, with an average power draw of 0.687 kW. The process utilized 1.94 kg of cutting fluid for cooling and lubrication throughout the machining cycle.

Table: Machining Operations - Build #13

Resource	Quantity
Electricity Consumption	2.84 kWh
Machining Time	4.133 hours
Average Power	0.687 kW
Cutting Fluid	1.94 kg

The machining time of 4.133 hours represents a 6.5% reduction compared to Build #12’s 4.42 hours, attributable to improved toolpath optimization and reduced need for intermediate measurements.

5.2 Material Removal Efficiency

The machining process removed 0.116 kg of material as chips from the 1 mm allowance, resulting in a finished propeller mass of 0.204 kg. This represents a material utilization efficiency of 63.8% from powder to final product, calculated from the initial 0.352 kg of powder consumed.

When compared to Build #12, which produced a 0.210 kg finished propeller from 0.380 kg of powder (55.3% utilization), the current build demonstrates significant improvement in material efficiency. The reduction in machining allowance from 1.2 mm in Build #12 to the current 1.0 mm standard has contributed substantially to this enhancement.

6. Output and Waste Stream Analysis

The production process yields both the finished product and various waste streams that require management and disposal.

6.1 Product Output

Build #13 produced one marine propeller with a final mass of 0.204 kg, meeting all dimensional and quality specifications. The component passed all non-destructive testing and met surface finish requirements without requiring rework.

Table: Production Output - Build #13

Output Type	Quantity
Marine Propeller	0.204 kg, 1 unit
Machined Chips	0.116 kg

The machined chips, consisting of ER70 steel, are collected for recycling, representing a recoverable material stream within our production ecosystem.

6.2 Waste Generation Trends

The 0.116 kg of machining chips from Build #13 represents a 22.7% reduction compared to Build #12’s 0.150 kg, directly resulting from the reduced machining allowance. This improvement has positive implications for both material costs and waste management operations.

No other significant waste streams were recorded during Build #13, with powder recovery systems effectively capturing and recycling unused material from the SLM process. Historical data shows consistent improvement in waste reduction, with Build #11 generating 0.180 kg of machining chips from a 1.5 mm allowance.

7. Integrated Performance Assessment

Synthesizing data across all process stages provides a comprehensive view of manufacturing efficiency and identifies opportunities for further optimization.

7.1 Total Energy Consumption

The complete manufacturing process for Build #13 consumed 15.158 kWh of electricity across all stages: 0.828 kWh for powder production, 11.49 kWh for SLM building, and 2.84 kWh for finish machining. This represents a specific energy consumption of approximately 74.3 kWh per kilogram of finished propeller.

When compared to Build #12's total energy consumption of 16.24 kWh (76.4 kWh/kg), the current build shows a 6.7% improvement in energy efficiency per unit mass. Industry references for similar SLM processes typically range from 70-85 kWh/kg for steel components, placing our performance within the competitive range.

7.2 Material Flow Efficiency

The complete material flow from initial billet to finished product demonstrates the interplay between process stages. Beginning with 0.414 kg of steel billet, the process yielded 0.352 kg of powder, which subsequently produced a 0.320 kg printed component before machining, ultimately resulting in the 0.204 kg finished propeller.

The overall material efficiency from billet to final product calculates to 49.3%, representing substantial improvement over Build #12's 42.1% efficiency. This enhancement primarily stems from reductions in both powder production losses and machining allowance.

8. Variance Analysis and Root Cause Investigation

Detailed examination of performance variations between builds identifies specific areas for process improvement and establishes correlation between operational parameters and resource consumption.

8.1 Energy Consumption Variances

The reduction in SLM electricity consumption from 12.5 kWh in Build #12 to 11.49 kWh in Build #13 correlates with a 15-minute reduction in build time despite identical component geometry. Investigation revealed that improved support structure design reduced recoating time by minimizing unnecessary powder deposition areas.

The consistency in powder production energy consumption (0.828 kWh vs. 0.850 kWh in Build #12) indicates stable operation of the atomization system, with minor variations attributable to ambient temperature effects on cooling efficiency.

8.2 Gas and Fluid Usage Patterns

Argon consumption reduction in the SLM process from 1.35 m³ to 1.2 m³ between builds resulted from implementation of improved chamber purge protocols that minimize gas waste during initial atmosphere establishment. The compressed air increase from 3.50 m³ to 3.56 m³ correlates with additional filtration cycles implemented to maintain powder quality standards.

Cutting fluid consumption decreased from 2.10 kg in Build #12 to 1.94 kg in Build #13, directly corresponding with the reduced machining time and material removal volume.

9. Recommendations for Process Optimization

Based on the comparative analysis between Build #13 and previous operations, the following recommendations are proposed to enhance manufacturing efficiency and resource utilization:

- 1. Standardize Improved Powder Production Parameters:** The 8.0% reduction in billet consumption achieved in Build #13 should be established as the new baseline, with procedures documented and implemented across all future production.
- 2. Optimize SLM Gas Management:** The successful argon reduction in Build #13 demonstrates that further gas savings may be achievable. Recommend conducting dedicated trials to determine minimum sufficient purge volumes without compromising part quality.
- 3. Extend Machining Allowance Reduction:** The positive results from reducing machining allowance from 1.2 mm to 1.0 mm suggest potential for further reduction. Propose experimental builds with 0.8 mm allowance to assess feasibility and impact on finishing requirements.
- 4. Implement Advanced Toolpath Strategies:** The 6.5% reduction in machining time between builds indicates that continued focus on CNC programming optimization can yield additional time and energy savings.
- 5. Establish Continuous Monitoring Protocols:** The detailed data collected for this analysis has proven invaluable. Recommend formalizing the data collection methodology and establishing automated reporting for key performance indicators.

10. Conclusion

The multi-build comparative analysis of ER70 steel marine propeller production via Selective Laser Melting has demonstrated consistent improvement in both material and energy efficiency. Build #13 represents our most efficient production cycle to date, with notable advancements in powder utilization, energy consumption during the SLM phase, and reduced waste generation from machining operations.

The distributed examination of process parameters across production stages has revealed interconnected relationships between operational decisions and resource consumption. The implementation of a 1 mm machining allowance has proven particularly effective in balancing finishing requirements with material conservation.

Continued focus on process optimization, informed by the detailed data analysis presented in this report, will enable further enhancements to our manufacturing efficiency while maintaining the high quality standards required for marine applications. The establishment of Build #13 as a performance benchmark provides a solid foundation for ongoing improvement initiatives.